



From Exotic Fruit to Premium Product – Dragon Fruit Value Addition



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Abstract

Dragon fruit (*Hylocereus spp.*) is a nutritionally rich and visually appealing tropical fruit with increasing global demand. Despite being mainly consumed fresh, its high perishability limits shelf life and utilization. Value addition through processing offers a practical approach to reduce post-harvest losses and enhance economic returns. The pulp can be processed into juices, ready-to-drink beverages, powders, dehydrated products, jams, jellies, candies, fruit leather, frozen desserts, and fermented products. In addition, the peel, often discarded as waste, is a valuable source of dietary fibre, antioxidants, and natural pigments, suitable for functional ingredients, natural colourants, and animal feed. Utilization of both pulp and peel supports improved nutrition, sustainability, and market expansion

Introduction

Dragon Fruit

Dragon fruit (*Hylocereus spp.*) is an emerging horticultural crop gaining popularity worldwide due to its attractive appearance, mild sweetness, and high nutritional value. Native to Central America, it is now widely cultivated in tropical and subtropical regions. In India, dragon fruit cultivation has expanded rapidly in recent years, with Gujarat, Maharashtra, Karnataka, Tamil Nadu, and Andhra Pradesh emerging as major producing states due to its adaptability, drought tolerance, and high market demand (FAO, 2021).

Dragon fruit offers numerous health benefits and is used in the prevention of various diseases. It supports eye health and improves the functioning of the kidney and brain, strengthens bones, and helps lower blood sugar, cholesterol, and the risk of colon cancer (Suryono, 2006). Its cholesterol-lowering effect is attributed to phytochemicals and plant sterols that regulate cholesterol metabolism, along with active compounds such as thiols, polyphenols, antioxidants, and betacyanin. Dragon fruit also aids digestion by detoxifying harmful substances like heavy metals and helps improve resistance to cough and asthma (Wybraniec and Mizrahi, 2002). Additionally, it promotes wound healing, prevents oxidative damage, and enhances probiotic growth in the intestinal tract (Zainoldin and Baba, 2009). Owing to its rich nutrient content, attractive appearance, and medicinal properties, dragon fruit is increasingly popular among health-conscious consumers.

Value Added Products from Dragon Fruit Pulp

Dragon Fruit Juice and Beverages

Juice Preparation

Dragon fruit juice is prepared by selecting fully ripe fruits, followed by washing, peeling, and pulp extraction. The pulp is crushed and filtered to remove seeds and fibrous matter, producing a smooth juice. The juice may be standardized by adding water, sugar, or natural acids to improve flavour balance. Pasteurization is carried

out to inactivate spoilage microorganisms and enzymes, thereby ensuring microbial safety and extending shelf life. The processed juice is then bottled or packed aseptically under hygienic conditions.

The resulting juice is visually appealing, especially from red-fleshed varieties, with a pleasant aroma and refreshing taste.

Ready-to-Drink (RTD) Beverages

Dragon fruit juice is widely used in the development of ready-to-drink beverages, which offer convenience and consistent quality. These include:

- Dragon fruit nectar: Prepared by blending a specific proportion of pulp with water and sugar, resulting in a thicker beverage with enhanced mouthfeel.
- Squash and concentrates: Highly concentrated products with added sugar and preservatives, diluted before consumption, and suitable for long-term storage.
- Blended beverages: Dragon fruit juice is often combined with citrus, pineapple, apple, or grape juice to enhance flavour, acidity, and nutritional value.

These products are popular in urban markets due to their attractive colour, refreshing nature, and ease of consumption.



Dragon Fruit Powder

Dragon fruit powder is prepared by drying the pulp and grinding it into a fine powder. It is used as a natural colourant, flavouring agent, and functional ingredient in bakery products, beverages, confectionery, instant mixes, and nutraceutical formulations.



Dehydrated Products

Sliced dragon fruit is dehydrated using hot air, solar, or vacuum drying to produce dried slices and chips. These products have extended shelf life and are consumed as healthy snacks or used as toppings in cereals and desserts.



Frozen and Confectionery Products

Dragon fruit pulp is used in ice creams, sorbets, candies, and gummies. The natural colour and mild flavour enhance visual appeal and consumer acceptance.



Dragon Fruit Jam

Jam is prepared by cooking dragon fruit pulp with sugar and acid until a suitable gel consistency is achieved. Due to its natural colour and mild flavour, dragon fruit jam is highly appealing and can be used as a spread or bakery filling. It has good storage stability when properly packed.

Dragon Fruit Jelly

Jelly is made from clarified dragon fruit juice with added pectin and sugar. The vibrant pink or red colour gives the jelly premium visual appeal. It is widely used in desserts and confectionery products.

Dragon Fruit Candy and Fruit Leather



Candy is prepared by osmotic dehydration of fruit pieces in sugar syrup, followed by drying. Fruit leather is made by spreading sweetened pulp into thin layers and drying. Both products are shelf-stable, nutritious, and popular among children.



Fermented Products

Dragon fruit can be used for the preparation of wine, vinegar, and probiotic beverages. Fermentation enhances shelf life, flavour, and functional properties, adding value to surplus or over-ripe fruits.

**Value Added Products from Dragon Fruit Peel**

Dragon fruit peel, often discarded as waste, constitutes a significant portion of the fruit and is rich in dietary fibre, pectin, polyphenols, and natural pigments. Its utilization supports waste reduction and sustainable processing.

Dragon Fruit Peel Powder

The peel is washed, blanched, dried, and ground into powder. This powder is rich in fibre and antioxidants and can be incorporated into bakery products, noodles, pasta, health mixes, and functional foods.

Natural Food Colorant

Red dragon fruit peel is a rich source of betalains, which can be extracted and used as a natural food colourant in beverages, dairy products, confectionery, and desserts, replacing synthetic dyes.

Bakery and Snack Products

Peel powder can be used to fortify cookies, bread, crackers, and extruded snacks, improving fibre content and antioxidant activity without significantly affecting sensory quality.

Pectin Extraction

Dragon fruit peel is a promising source of pectin, which can be extracted and used as a gelling and stabilizing agent in jams, jellies, and pharmaceutical formulations.

Animal Feed and Compost

Dried peel can be utilized as livestock feed or organic manure, contributing to waste valorisation and circular economy.

Conclusion

Dragon fruit offers significant potential for value addition through diversified processed products that enhance shelf life, marketability, and income generation. Processing of the pulp into beverages, preserves, snacks, and fermented products, along with effective utilization of the peel for functional and feed applications, promotes waste reduction and sustainability. Strengthening value addition practices can support farmers, reduce post-harvest losses, and meet the growing demand for healthy and functional food products, making dragon fruit a promising crop for the food processing industry.

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